

## Suyoung Lee

---

|                     |                                                                                                                                                                                                                         |                                                                                                                                                                                  |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CONTACT INFORMATION | 231 Teheran-ro, Gangnam District,<br>Seoul, Republic of Korea                                                                                                                                                           | <b>Email:</b> <a href="mailto:suyounglee424@gmail.com">suyounglee424@gmail.com</a><br><b>Homepage:</b> <a href="https://suyoung-lee.github.io">https://suyoung-lee.github.io</a> |
| RESEARCH INTERESTS  | AI Agent<br>Agent harnessing, automated research, reasoning, and planning.<br>Reinforcement Learning – practical application<br>Meta-RL, generalization, sample efficiency, exploration, and offline RL.                |                                                                                                                                                                                  |
| CAREER              | <b>Krafton</b><br>Gameplay AI Team, AI Research Division, Seoul.<br>- Large-scale distributed RL and VLA training on PUBG                                                                                               | <i>Aug. 2025 – present</i>                                                                                                                                                       |
|                     | <b>Samsung Electronics</b><br>Semantic T/F, AI Model Team, Samsung Research, Seoul.<br>- Finetuning on-device LLM: screen entity extraction.<br>- AI Agent Framework: GUI Agent and Text-to-SQL Agent.                  | <i>Mar. 2024 – Jul. 2025</i>                                                                                                                                                     |
| EDUCATION           | <b>Ph.D.</b> , Electrical Engineering.<br>Korea Advanced Institute of Science and Technology (KAIST).<br>Advisor: Prof. Youngchul Sung.                                                                                 | <i>Aug. 2022 – Feb. 2024</i>                                                                                                                                                     |
|                     | <b>Ph.D. Candidate</b> , Electrical Engineering.<br>Korea Advanced Institute of Science and Technology (KAIST).<br>Advisor: Prof. Sae-Young Chung.                                                                      | <i>Feb. 2019 – Aug. 2022</i>                                                                                                                                                     |
|                     | <b>M.S.</b> , Electrical Engineering.<br>Korea Advanced Institute of Science and Technology (KAIST).<br>Advisor: Prof. Sae-Young Chung.                                                                                 | <i>Feb. 2017 – Feb. 2019</i>                                                                                                                                                     |
|                     | <b>B.S.</b> , Electrical Engineering<br>Korea Advanced Institute of Science and Technology (KAIST).                                                                                                                     | <i>Feb. 2012 – Feb. 2017</i>                                                                                                                                                     |
|                     | Hansung Science High School, Seoul, Republic of Korea.                                                                                                                                                                  | <i>Mar. 2010 – Feb. 2012</i>                                                                                                                                                     |
| HONORS              | <b>Outstanding Ph.D. Dissertation Award.</b><br>Thesis: <i>Meta-Reinforcement Learning with Imaginary Tasks</i> , KAIST EE.                                                                                             | <i>2024</i>                                                                                                                                                                      |
|                     | Qualcomm-KAIST Innovation Awards.<br>Paper competition awards for graduate students, Qualcomm.                                                                                                                          | <i>2018</i>                                                                                                                                                                      |
|                     | Un Chong-Kwan Scholarship Award.<br>For achievement of excellence in 2017 entrance examination, KAIST EE.                                                                                                               | <i>2017</i>                                                                                                                                                                      |
| PUBLICATIONS        | [C] Conference    [W] Workshop    [P] Preprint    [J] Journal                                                                                                                                                           |                                                                                                                                                                                  |
|                     | [C1] <b>Suyoung Lee</b> , Sungik Choi, and Sae-Young Chung. “ <i>Sample-Efficient Deep Reinforcement Learning via Episodic Backward Update</i> .” Neural Information Processing Systems (NeurIPS) 2019.                 |                                                                                                                                                                                  |
|                     | [C2] <b>Suyoung Lee</b> and Sae-Young Chung. “ <i>Improving Generalization in Meta-RL with Imaginary Tasks from Latent Dynamics Mixture</i> .” Neural Information Processing Systems (NeurIPS) 2021.                    |                                                                                                                                                                                  |
|                     | [C3] <b>Suyoung Lee</b> , Myungsik Cho, and Youngchul Sung. “ <i>Parameterizing Non-Parametric Meta-Reinforcement Learning Tasks via Subtask Decomposition</i> .” Neural Information Processing Systems (NeurIPS) 2023. |                                                                                                                                                                                  |

- [C4] Jeonghye Kim, **Suyoung Lee**, Woojun Kim, and Youngchul Sung “*Decision ConvFormer: Local Filtering in Metaformer is Sufficient for Decision Making.*” International Conference on Learning Representations (ICLR) 2024 as **spotlight presentation (366/7262=5.0%)**.
- [C5] Myungsik Cho, Jongeui Park, **Suyoung Lee**, and Youngchul Sung. “*Hard Task First: Multi-Task Reinforcement Learning through Task Scheduling.*” International Conference on Machine Learning (ICML) 2024.
- [C6] Jeonghye Kim, **Suyoung Lee**, Woojun Kim, and Youngchul Sung “*Adaptive Q-Aid for Conditional Supervised Learning in Offline Reinforcement Learning.*” Neural Information Processing Systems (NeurIPS) 2024.
- [W1] **Suyoung Lee** and Sae-Young Chung. “*Adaptive Intrinsic Motivation with Decision Awareness.*” Decision Awareness in Reinforcement Learning Workshop at International Conference on Machine Learning (ICML) 2022.
- [W2] Jeonghye Kim, **Suyoung Lee**, Woojun Kim, and Youngchul Sung. “*Decision ConvFormer: Local Filtering in MetaFormer is Sufficient for Decision Making.*” Foundation Models for Decision Making Workshop at Neural Information Processing Systems (NeurIPS) 2023.
- [J1] Gyeongmin Kim, Jeonghye Kim, **Suyoung Lee**, Jaewoo Baek, Howon Moon, Sangheon Shin, and Youngchul Sung. “*Robust Reinforcement Learning Under Dimension-Wise State Information Drop.*” IEEE Access, 2024.

|                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |           |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| OPEN SOURCE              | <b>Prompt2Policy</b><br><a href="https://github.com/krafton-ai/Prompt2Policy">github.com/krafton-ai/Prompt2Policy</a>   <a href="#">blog</a><br><ul style="list-style-type: none"> <li>• Agentic engineering for reinforcement learning: describe a behavior in a prompt, get a trained policy.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                      | Apr. 2026 |
| LANGUAGES                | Korean (native)<br>English (fluent) – TOEIC 950 (23.06.28), OPIc IH (24.06.08)<br>Russian (basic)<br>(International linguistic experience at Tashkent International School, 2007–2009.)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |           |
| PROGRAMMING<br>LANGUAGES | Claude Code<br><del>MATLAB and Python (PyTorch/TensorFlow).</del><br>GitHub: <a href="https://github.com/suyoung-lee">https://github.com/suyoung-lee</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |           |
| ACADEMIC<br>SERVICES     | Invited talk <ul style="list-style-type: none"> <li>• KAIST EE Graduate School REEsearch Party: academic seminar by doctoral graduates who won outstanding thesis awards, April 2024.</li> </ul> Conference reviewer <ul style="list-style-type: none"> <li>• International Conference on Machine Learning (ICML): 2021–2025</li> <li>• Neural Information Processing Systems (NeurIPS): 2021–2025</li> <li>• International Conference on Learning Representations (ICLR): 2024</li> </ul> Program committee <ul style="list-style-type: none"> <li>• Foundation Models for Decision Making Workshop (FMDM) at Neural Information Processing Systems (NeurIPS) 2023.</li> </ul> |           |

TEACHING  
EXPERIENCE

Teaching assistant (KAIST)

*Spring 2018 to Fall 2020*

- EE326 Introduction to Information Theory and Coding.
- EE210 Probability and Introductory Random Processes.
- EE105 Electrical Engineering: Changing the World.
- EE405 Electronics Design Lab. Network of Smart Things.
- EE807 Special Topics in EE. Deep Reinforcement Learning and AlphaGo.
  - Course rewarded for the **outstanding TA award** at KAIST EE.
- EE405 Electronics Design Lab. Network of Smart Systems.